

Egyptian Hieroglyph Classification Using Fine-Tuned EfficientNet-B3: A Deep Learning Approach

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Abstract

Ancient Egyptian hieroglyphs represent one of humanity's most significant writing systems, with millions of inscriptions still awaiting full decipherment across temples, tombs, papyri, and everyday artefacts. Automated recognition is critical not merely for accelerating translation, but for preserving inscriptions threatened by erosion and recovering knowledge spanning medicine, astronomy, and engineering that predates classical Western texts. Beyond scholarship, this work recovers authentic human voices: artisans, physicians, and ordinary people whose recorded thoughts retain striking resonance today. We present a deep learning classification system for ancient Egyptian hieroglyphs using a fine-tuned EfficientNet-B3 model trained on the merged Glyph2025 dataset a combination of GlyphHiero2025 and GlyphFranken2025 encompassing 353 Gardiner sign classes across 13,742 images. Training employs a two-phase gradual unfreeze strategy, class-balanced augmentation, and weighted random sampling to counter severe class imbalance. The proposed model achieves 91.97% Top-1 accuracy, 96.38% Top-3, 97.30% Top-5, and a Weighted F1 score of =0.9048 the broadest single-model class coverage reported in this field.

CCS Concepts

• **Computing methodologies** → **Image classification; Computer vision; Neural networks**; • **Applied computing** → **Arts and humanities**.

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Keywords

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1 Introduction

Ancient Egyptians used hieroglyphs for over 3,000 years, developing one of history's most visually intricate and symbolically rich writing systems. Even after the discovery of the Rosetta Stone, scholars required more than two decades to decode their underlying logic a testament to the system's extraordinary complexity. Today, millions of inscriptions survive on temple walls, tomb ceilings, papyri, and everyday objects, yet vast numbers remain only partially deciphered [7]. The sheer volume of surviving material far outpaces what human experts can realistically process, making automated recognition not merely a convenience but a genuine necessity for the field.

Hieroglyph classification presents a fine-grained visual recognition challenge of unusual difficulty. The Gardiner sign list comprises over 750 distinct categories, many of which share striking visual similarities. A subtle shift in orientation or posture can entirely alter a sign's meaning the angle of a bird's wing or the position of a seated figure, for instance, carries significant semantic weight. Compounding this, the same sign can vary considerably depending on when, where, and by whom it was carved, as scribal conventions shifted substantially across dynasties and regions [2, 12, 13].

The stakes extend beyond academic interest. Hieroglyphic texts contain medical formulae, astronomical observations, agricultural

records, and funerary rites knowledge that illuminates early human civilisation. Digitisation programmes worldwide are accumulating high-resolution imagery of inscriptions faster than experts can annotate them. Reliable automated classification is therefore an essential pre-processing step for any downstream translation or retrieval pipeline.

Early approaches to automated recognition relied on hand-crafted features and conventional machine learning pipelines [1, 7]. Deep learning transformed the landscape considerably, with convolutional neural networks achieving competitive accuracy on curated benchmarks [2]. Subsequent work extended these capabilities to sign detection and spatial localisation [9, 15], and more ambitious systems combined recognition with transformer-based translation [3, 19] and retrieval-augmented language models [8].

Despite this progress, significant structural limitations persist. Most existing systems cover fewer than 200 Gardiner sign classes [2, 10], and available datasets remain small and stylistically narrow. Class imbalance continues to constrain generalisation [9, 13], while models trained under controlled conditions frequently underperform when applied to real inscription data [12, 13].

This paper directly addresses these gaps. We introduce Glyph2025, a merged dataset combining GlyphHiero2025 and GlyphFranken2025, spanning 353 Gardiner sign classes across 13,742 images. Leveraging a fine-tuned EfficientNet-B3 architecture with gradual unfreezing, targeted augmentation, and weighted random sampling, our system achieves state-of-the-art results.

The main contributions of this study:

- A unified 353 class hieroglyph benchmark the largest single-model class coverage reported to date by integrating GlyphHiero2025 and GlyphFranken2025 at 300×300 resolution.
- A two-stage fine-tuning strategy with differential learning rates and cosine annealing, enabling stable convergence under severe class imbalance.
- Achieved 91.97% Top-1, 96.38% Top-3, and 97.30% Top-5 accuracy, with a Weighted F1-score of 0.9048 across 353 classes, without ensemble techniques.
- A systematic analysis of failure modes, revealing that errors primarily occur between visually similar intra-group classes, highlighting the need for sequence-context modelling.

2 Related Work

Automated Egyptian hieroglyph recognition has been explored from multiple angles over the past decade. Early work by Duque-Domingo et al. [7] demonstrated real-time glyph identification for museum navigation using SVM+HOG on 30 classes. Barucci et al. [2] established CNNs as the standard approach, reaching 87.4% on 172 Gardiner classes from the Franken corpus. Lombardi et al. [15] extended this to object detection with YOLOv8, enabling localisation within multi-glyph scenes rather than isolated symbol recognition showing that per-sign accuracy often depends critically on correct segmentation upstream.

Dataset diversity has emerged as a central concern. Golyadkin et al. [12] showed that classification accuracy drops sharply across

writing styles, and their follow-up MEH benchmark [13] was designed specifically to stress-test models under palaeographic variation. Fuentes-Ferrer et al. [9] addressed structured inscription reading through segmentation-based voting, while Fuentes-Jimenez et al. [10] showed that specialised corpora like Coffin Texts require targeted fine-tuning beyond standard transfer learning.

On the translation side, De Cao et al. [3] adapted transformer architectures to Middle Egyptian, and the Context-Aware RAG system [8] combined SAM segmentation, fine-tuned InceptionV3, and retrieval-augmented generation for contextually grounded translation. Chen et al., through LogogramNLP [4], created a cross-lingual NLP benchmark for standardised comparison across ancient writing systems. End-to-end pipelines have also become popular: Nasser et al. [16] integrated segmentation, Gardiner mapping, and CNN-based translation in a single framework, while Agha et al. [1] jointly addressed detection, classification, and transliteration as early as 2018.

Class imbalance mitigation has received limited dedicated attention. Standard over-sampling and augmentation strategies are widely applied, but none of the surveyed systems combine offline augmentation with weighted random sampling at the scale of 353 classes. No prior work constructs a merged multi-source benchmark at this scale; the present study fills this gap directly. Table 1 summarizes these prior systems alongside our approach, highlighting the substantial gap in class coverage between Glyph2025 and existing work.

Table 1: Comparative Summary of Prior Work vs. Glyph2025 (Ours)

Study	Arch.	Cls	Dataset	Top-1
Barucci et al. [2]	CNN	172	Franken	87.40%
Duque-Domingo et al. [7]	SVM+HOG	30	Custom	85.20%
Golyadkin et al. [12]	ResNet-50	172	MEH	89.10%
Golyadkin et al. [13]	EfficientNet	172	MEH	91.30%
Lombardi et al. [15]	YOLOv8	50	Custom	88.60%
Ours	EffNet-B3	353	Glyph2025	91.97%

3 Methodology

This section covers the full pipeline used to build and evaluate the Egyptian hieroglyph classification system, from raw data collection through to final test-set evaluation. Figure 1 shows the full pipeline, from data preparation through preprocessing, training, and evaluation.

3.1 Computational Setup

All training was carried out on Kaggle using an NVIDIA Tesla T4 GPU with 15.6 GB of video memory. The code was written in PyTorch 2.x with mixed-precision training (FP16) enabled throughout to reduce memory usage and accelerate training without sacrificing numerical accuracy. Every random seed was fixed to 42 for reproducibility.

3.2 Dataset Preparation

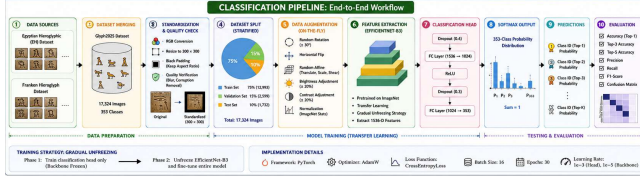


Figure 1: Classification Pipeline of the Proposed Fine-Tuned EfficientNet-B3 Model for 353-Class Egyptian Hieroglyph Recognition

The dataset used in this work is Glyph2025, built by combining two existing hieroglyph image collections. GlyphHiero2025 contributes 9,703 images covering 310 Gardiner sign classes from carved and painted stone inscriptions. GlyphFranken2025 adds 4,039 images across 171 classes drawn from inscriptions inside the Pyramid of Unas. Merging required lowercase-normalisation of all folder names to resolve naming inconsistencies between the two sources and avoid double-counting classes. The resulting unified corpus contains 13,742 images across 353 Gardiner sign classes, to our knowledge the broadest class coverage attempted with a single classification model in this field. The per-class distribution is highly skewed: minimum 1 image, maximum 987, median 7, reflecting real-world scarcity of rare signs.

Of the 353 total classes, 128 Gardiner codes are shared between GlyphHiero2025 and GlyphFranken2025, 182 classes are unique to GlyphHiero2025, and 43 are unique to GlyphFranken2025. For shared classes, images from both sources were pooled under a single label after the normalisation step described above; we found no conflicting Gardiner-code assignments between the two sources for any shared class. The two datasets differ visually: GlyphHiero2025 draws from a wider range of carved and painted stone surfaces with variable lighting and surface wear, whereas GlyphFranken2025 comes entirely from the Pyramid of Unas and so has more consistent stone type, carving style, and photographic conditions. Shared classes can therefore contain two visually distinct sub-populations, which may contribute to the confusion between visually similar signs discussed in Section 4.4. We found no image-level duplicates between the two sources, since each comes from a physically distinct set of inscriptions.

3.3 Image Preprocessing

Every image was resized to 300×300 pixels with zero-padding to preserve the original aspect ratio, since distorting hieroglyphic shapes could affect classification. All images were converted to RGB. Pixel values were normalised using ImageNet channel statistics:

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}, \quad \mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225] \quad (1)$$

This matches EfficientNet-B3’s native 300×300 input resolution exactly, eliminating any need for rescaling at inference time.

3.4 Dataset Splitting

Data was split using a per-class stratified approach with random seed 42, assigning 75% to training, 15% to validation, and 10% to test. Classes with fewer than four images were handled case-by-case to ensure all 353 classes appeared in the training set. Final counts: **10,227 train / 2,108 val / 1,407 test** images. The validation and test sets were never touched by augmentation.

3.5 Handling Class Imbalance

The dataset is heavily imbalanced smallest class 1 image, largest 987, median 7. Left uncorrected, the model would simply learn to favour the most common classes. We tackle this in two complementary ways.

Offline augmentation tops up any class with fewer than 50 training images to exactly 50 using combinations of rotation ($\pm 25^\circ$), horizontal and vertical flipping, Gaussian blur, auto-contrast, and random cropping. This expanded the training set from 10,227 to **23,107 images**. **Weighted random sampling** during training assigns each image a weight inversely proportional to its class frequency, so rare categories appear proportionally more often per batch, preventing the optimiser from devoting the majority of gradient signal to the most common 50 classes.

We chose 50 as a practical middle ground: a lower threshold leaves rare classes with too little signal, while a higher one pads them almost entirely with near-duplicate synthetic images. To check this, we retrained at thresholds of 30 and 100 with all else held fixed. Validation accuracy rose slightly with the higher threshold (92.03% $\rightarrow$ 92.17%), but test-set performance peaked at 50 — the highest Top-1 (91.97%), Macro F1 (0.6939), and Weighted F1 (0.9048) of the three, with Macro F1 dropping to 0.6879 at threshold=100. Extra augmentation beyond ~50 images per class does not appear to help real generalisation, so we keep 50 as the reported setting. Table 2 reports the full results across all three thresholds tested.

Table 2: Augmentation threshold sensitivity analysis.

Thr.	Val Acc	Top-1	Top-3	Top-5	Macro F1	Wtd F1
30	92.08%	91.47%	96.16%	96.94%	0.6950	0.8991
50 (rep.)	92.03%	91.97%	96.38%	97.30%	0.6939	0.9048
100	92.17%	91.47%	96.38%	96.87%	0.6879	0.8986

3.6 Online Data Augmentation

In addition to offline augmentation, random transformations were applied to each training image on the fly. Validation and test images received only resizing and normalisation. Table 3 summarises the parameters used.

3.7 Model Architecture

We used EfficientNet-B3 as the backbone, initialised with ImageNet-pretrained weights. This model was chosen specifically because its default input size of 300×300 pixels matches our preprocessing resolution exactly. EfficientNet scales depth, width, and resolution jointly via compound scaling, giving a strong balance between

Table 3: Online augmentation parameters used during training.

Augmentation	Parameter
Random Resized Crop	Scale 0.65–1.0
Rotation	$\pm 25^\circ$
Horizontal Flip	$p = 0.5$
Vertical Flip	$p = 0.2$
Colour Jitter	Brightness, Contrast, Saturation
Random Grayscale	$p = 0.15$
Random Erasing	$p = 0.25$

accuracy and computational cost relative to alternatives such as ResNet-50 or DenseNet-169.

The backbone produces a 1,536-dimensional feature vector via global average pooling:

$$f(x) = \phi_{\text{EMB3}}(x), \quad f(x) \in \mathbb{R}^{1536} \quad (2)$$

The original classification head is replaced with a custom head consisting of: Dropout($p=0.4$) $\rightarrow$ FC(1536 $\rightarrow$ 1024) $\rightarrow$ ReLU $\rightarrow$ Dropout($p=0.3$) $\rightarrow$ FC(1024 $\rightarrow$ 353). The final prediction is:

$$\hat{y} = \text{Softmax}(W_2 \text{ReLU}(W_1 f(x))) \quad (3)$$

The stacked dropout layers provide aggressive regularisation given the limited data per class. The full architecture from input through MBConv blocks to the custom head is illustrated in Fig. 2.

3.8 Training Configuration

Training proceeds in two phases. In **Phase 1** (epochs 1–5), the backbone was completely frozen and only the new classification head was trained using AdamW with a learning rate of 1×10^{-3} following a cosine schedule decaying to 1×10^{-4} . These epochs allowed the randomly-initialised head to stabilise before backbone weights were perturbed.

In **Phase 2** (epochs 6–80), the entire network was unfrozen with differential learning rates 3×10^{-5} for the backbone and 1×10^{-4} for the head preceded by a 3-epoch linear warm-up, then cosine annealing:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos \frac{t\pi}{T}) \quad (4)$$

The lower backbone rate prevents catastrophic forgetting of ImageNet representations while the higher head rate allows rapid domain adaptation. Cross-entropy loss with label smoothing ($\epsilon=0.1$, $K=353$) was applied:

$$\mathcal{L} = -(1-\epsilon) \log p_y - \frac{\epsilon}{K} \sum_{i=1}^K \log p_i \quad (5)$$

AdamW weight decay 10^{-4} , gradient clipping at norm 1.0, batch size 16. Early stopping triggered after 15 epochs without validation improvement; the best checkpoint was saved at epoch 58 out of 73 total epochs.

3.9 Evaluation Protocol

The saved best checkpoint was evaluated on the held-out test set. We measured Top-1, Top-3, and Top-5 accuracy, along with Macro F1 and Weighted F1. Top- k metrics are particularly relevant here because in practical use a hieroglyph recognition tool would likely

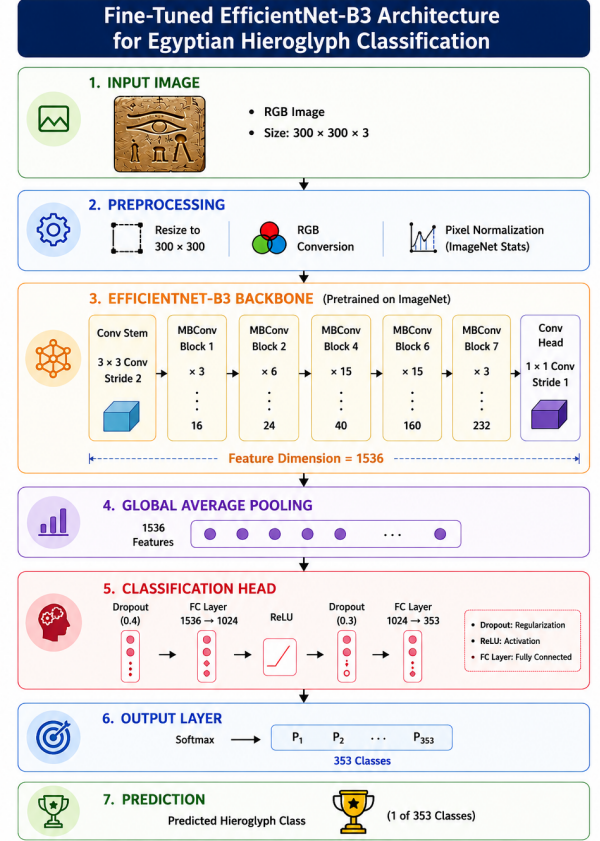


Figure 2: Fine-tuned EfficientNet-B3 architecture for Egyptian hieroglyph classification. Input: 300×300 RGB image. Backbone: Conv stem $\rightarrow$ MBConv blocks 1–7 $\rightarrow$ Conv head, producing 1,536-dim features via global average pooling. Custom head: Dropout(0.4) $\rightarrow$ FC(1536 $\rightarrow$ 1024) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ FC(1024 $\rightarrow$ 353) $\rightarrow$ Softmax over 353 classes.

present an expert with several candidate predictions rather than committing to one. A full 353×353 confusion matrix was generated in both raw and row-normalised form, and per-class accuracy was computed for every Gardiner sign category individually.

4 Results

4.1 Training Dynamics

The model ran for 73 epochs before early stopping triggered. Phase 1 (frozen backbone, epochs 1–5) raised training/validation accuracy from 20.4%/11.0% to 48.0%/30.3% establishing a stable head initialisation without disturbing the pretrained features.

When Phase 2 opened the full backbone at epoch 6, the response was immediate: epoch 7 alone saw training accuracy jump to 65.8% and validation to 52.0%. Training crossed 90% at epoch 13; validation peaked at **92.03%** at epoch 58 with no further improvement in the remaining 15 epochs. Training loss fell from 4.54 to ≈ 1.02 over the full run (Fig. 3). The training and validation curves track each

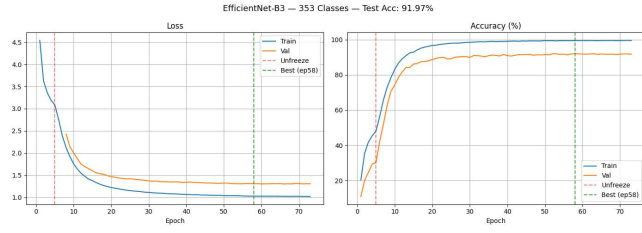


Figure 3: Training and validation loss (left) and accuracy (right) over 73 epochs. The red vertical line marks the Phase 1→Phase 2 transition at epoch 5. The green vertical line marks the best checkpoint at epoch 58 (val acc = 92.03%). The sharp acceleration at epoch 6 confirms effective transfer of ImageNet features.

other closely, indicating no severe overfitting despite very small per-class sample sizes.

4.2 Test Set Performance

Table 4: Test set results for the proposed EfficientNet-B3 classifier on Glyph2025 (353 classes, $n=1,407$ images).

Metric	Score
Top-1 Accuracy	91.97%
Top-3 Accuracy	96.38%
Top-5 Accuracy	97.30%
Weighted F1	0.9048
Macro F1	0.6939

The model achieved 91.97% Top-1 on the held-out test set (Table 4). The Top-3 and Top-5 rates (96.38% and 97.30%) confirm that the correct class is nearly always within the model’s top few predictions, practically valuable for expert-assisted tools where a short candidate list is presented for human verification. The Weighted F1 of 0.9048 reflects strong performance on well-represented classes. The lower Macro F1 (0.6939) is driven by rare categories with a single test image, where one wrong prediction yields an F1 of zero, a data coverage issue rather than a model failure. We note that this also means per-class performance on rare Gardiner signs is not reliably measured: a single test image cannot establish whether the model generalises to that class or was simply correct or incorrect on one sample. Reported accuracy for these classes should therefore be treated as a coverage gap rather than a validated measurement of the model’s true rare-class performance.

4.3 Explainability

To examine what the model is learning, we apply Gradient-weighted Class Activation Mapping (Grad-CAM) to the final convolutional block of the fine-tuned EfficientNet-B3. Grad-CAM produces a coarse localization map of the image regions most influential to the model’s prediction, which lets us check whether the classifier responds to the glyph strokes themselves rather than background texture, stone damage, or preprocessing artifacts.

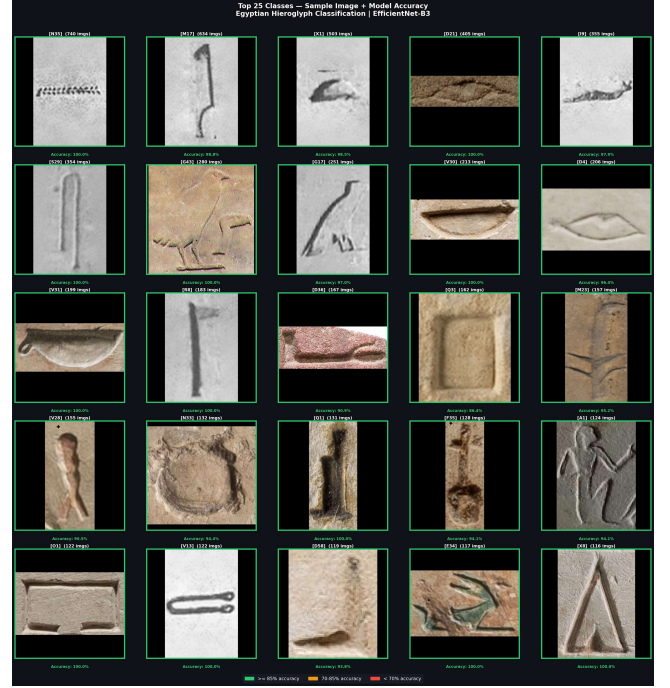


Figure 4: Sample images from the top 25 most represented training classes with their test accuracy. Green borders: $\geq 85\%$; orange: 70–85%; red: $< 70\%$. All top-25 classes achieve high accuracy, confirming that well-represented categories are reliably classified.

Figure 5 shows Grad-CAM overlays for test images from both well-represented and rare classes. In most cases the activation concentrates on the glyph’s distinguishing strokes, for example the curved beak of a bird glyph or the enclosed loop of a cartouche-like symbol. This is consistent with the model learning morphological features of the hieroglyphs rather than incidental cues such as stone color or lighting. In a small number of misclassified examples, activation spreads across the glyph boundary or shifts toward damaged or occluded regions of the source image, which suggests preprocessing quality contributes to the residual error rate.

4.4 Per-Class Performance

Of the 257 classes represented in the test set, **171** attained $\geq 85\%$ accuracy, 10 fell in the 70–85% band, and 76 fell below 70%. The mean per-class accuracy was 72.41% but the **median was 100%** most classes are classified perfectly; a small group of hard cases with minimal test coverage drags the mean down (Fig. 6). Figure 4 shows sample images from the 25 most represented classes, all of which achieve strong test accuracy.

The 10 lowest-F1 classes (a17, a26, a28, a9, aa11, aa27, aa5, d19, d38, d41) each had exactly one test image. Ten classes achieved perfect F1 = 1.000 (w17, w5, w9, x7, x8, z1, z11, z4, z7, z9) — predominantly geometric marks in the Z and X groups that are visually distinctive and thus reliably classified.

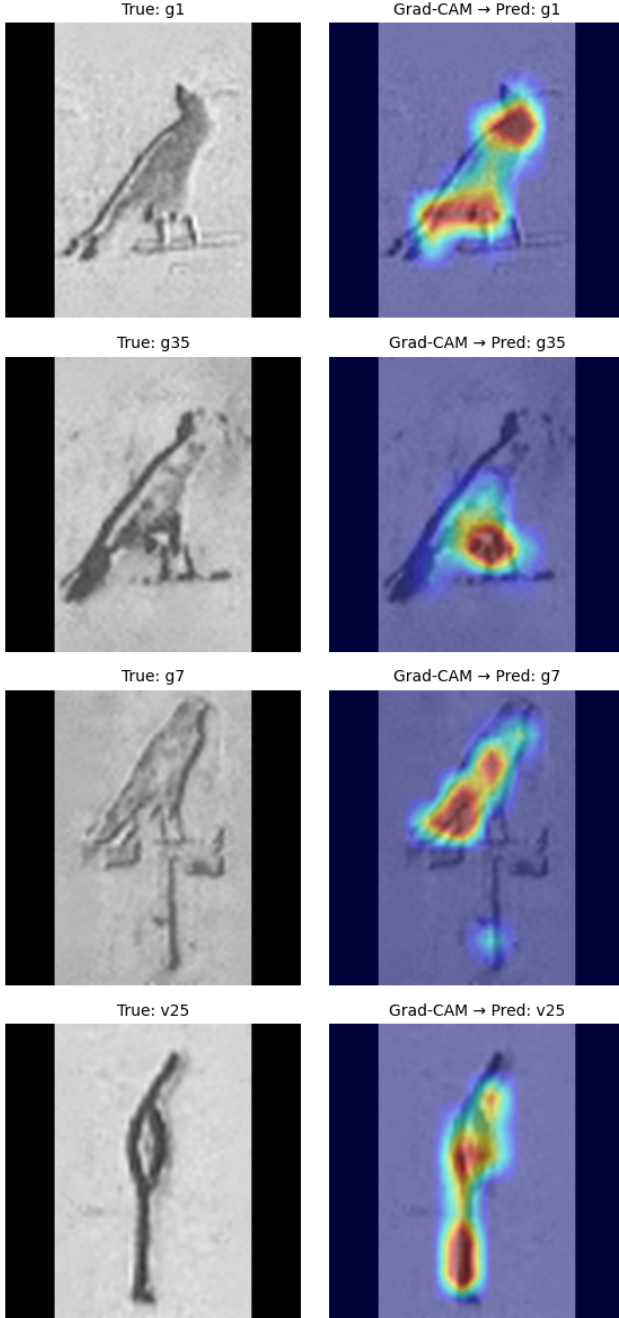


Figure 5: Grad-CAM visualizations for representative test images. Warmer colors (red/yellow) indicate regions of highest influence on the predicted class; cooler colors (blue) indicate low influence. The model consistently attends to the distinguishing strokes of each glyph.

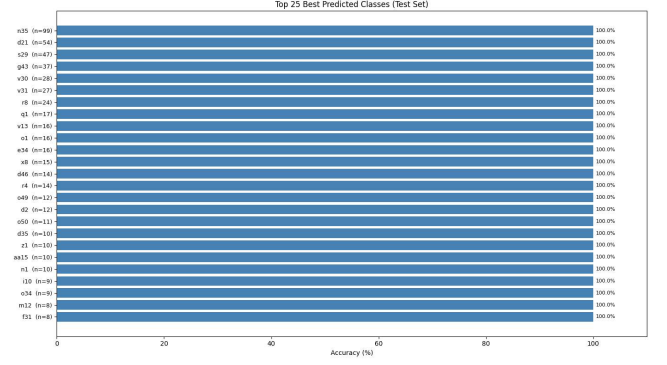


Figure 7: Top 25 best-predicted Gardiner sign classes on the test set, ranked by number of test samples. All 25 classes achieve 100% Top-1 accuracy, with sample counts ranging from $n=8$ (f31, m12) to $n=99$ (n35).

4.5 Confusion Analysis

Figure 7 shows the 25 best-predicted Gardiner sign classes on the test set, all of which reach perfect Top-1 accuracy. Fig. 8 shows the confusion matrix for the 25 most represented test classes. The diagonal is strongly dominant. The worst confusion was g1 predicted as g35 twice both are depictions of seated male figures differing only in subtle positional details. Every other confused pair occurred exactly once across 1,407 test images.

Errors cluster between signs within the same Gardiner alphabetic group (A-group human figures, G-group birds, D-group body parts), which is precisely the expected failure mode for a purely visual classifier without sequence context. This pattern strongly motivates future work integrating grammatical or positional constraints from surrounding inscriptions.

5 Discussion

5.1 Impact of the Two-Phase Training Strategy

Reaching 91.97% Top-1 over 353 Gardiner classes nearly double the class count of most prior systems [2, 10] demonstrates that a merged dataset with a careful training regime is sufficient to extend coverage substantially without resorting to ensemble methods. The two-phase gradual unfreeze played a visible role: the 5 frozen-backbone epochs gave the new head time to stabilise before the ImageNet features were released, and the immediate jump from 30.3% to 52.0% validation accuracy at epoch 6 confirms the backbone’s representations transferred effectively [12, 13].

Differential learning rates (3×10^{-5} backbone, 1×10^{-4} head) proved critical. Using a uniform learning rate for both layers caused validation oscillation in preliminary experiments, eventually recovering to similar accuracy but requiring approximately 20 additional epochs. Cosine annealing prevented sharp loss spikes near the end of training that might otherwise dislodge a well-converged head solution.

5.2 Understanding the F1 Divergence

The Weighted/Macro F1 divergence (0.9048 vs. 0.6939) reflects thin test coverage for rare signs, not a fundamental model limitation a

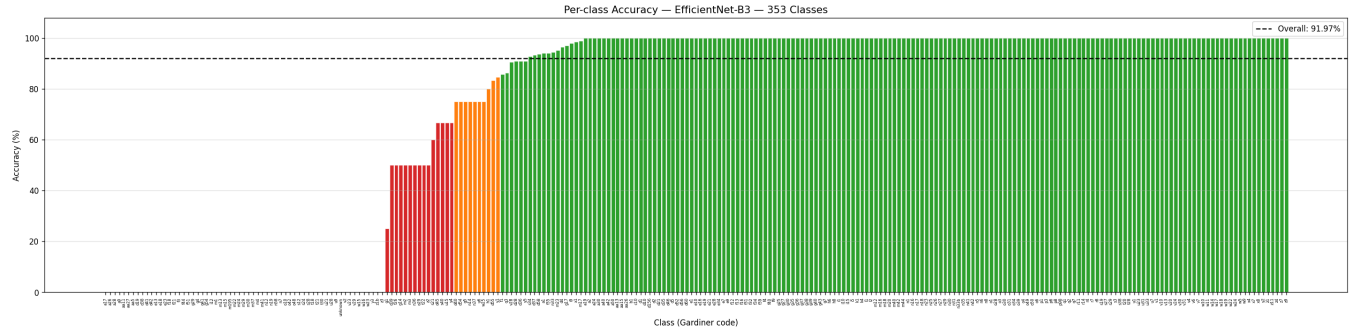


Figure 6: Per-class accuracy across all 257 evaluated test classes, sorted in ascending order. Green bars $\geq 85\%$ (171 classes), orange 70–85% (10 classes), red $< 70\%$ (76 classes). The long green plateau on the right shows that the majority of classes are classified perfectly; low-accuracy cases are almost exclusively those with a single test image.

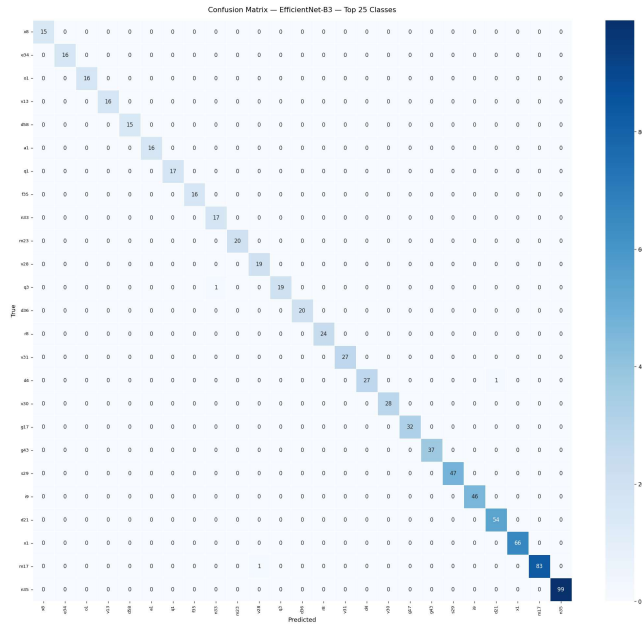


Figure 8: Confusion matrix for the top 25 most represented classes in the test set. The diagonal strongly dominates. Worst-case confusion: g1 predicted as g35 (twice); both depict seated male figures differing only in subtle posture details. All other off-diagonal entries are at most 1 across 1,407 test images.

point underscored by the 100% median per-class accuracy and the 171 classes above 85%. Addressing this requires more images for rare Gardiner categories, either through new digitisation or synthetic generation [11, 17]. We estimate that raising the minimum class size from 1 to 10 test images across all categories would likely bring Macro F1 within 5 percentage points of the Weighted figure [14, 18].

5.3 Limitations and Failure Modes

Both source datasets primarily contain isolated or semi-synthetic glyphs rather than photographic crops from real inscriptions. As a result, we did not validate the model on photographs of real, uncurated inscriptions, and performance on in-situ imagery of temple walls, tomb ceilings, or artefacts remains untested. Cross-style generalisation [12, 13] and scene-level detection remain open challenges. The confusion pattern errors almost exclusively between visually similar signs within the same Gardiner group suggests that incorporating sequence context (as in [3, 8]) or attention-based architectures could further reduce misclassifications. Hyperspectral imaging and multimodal documentation [5, 6] also open complementary avenues for improving inscription-level accuracy.

5.4 Pathways to 750+ Classes

Glyph25 is a step forward in raw class coverage; the logical next goal is expanding toward the full 750+ Gardiner list, where data scarcity is increasingly severe [2, 9]. Three complementary strategies emerge from our analysis: (i) active digitisation of archival photograph collections targeting rare sign types; (ii) generative augmentation with diffusion models conditioned on glyph structure [11, 17]; and (iii) cross-dataset transfer from hieratic and demotic scripts sharing visual roots with certain Gardiner signs. Stronger classification also feeds directly into downstream translation pipelines [3, 19] and retrieval-augmented systems [8], with broader NLP benchmarking efforts on logographic languages providing complementary insights [4].

6 Conclusion and Future Work

This work constructed **Glyph25** a 353-class, 13,742-image merged Egyptian hieroglyph benchmark and trained a fine-tuned EfficientNet-B3 with two-phase gradual unfreezing, offline augmentation, and weighted random sampling. The system achieves **91.97%** Top-1, **96.38%** Top-3, **97.30%** Top-5 accuracy and a Weighted F1 of **0.9048** to our knowledge the widest single-model Gardiner-class coverage reported, building on earlier integrated detection and transliteration pipelines [1]. Most classes are classified perfectly; the Macro F1 of 0.6939 reflects rare categories where a single wrong guess

on a one-image test class wipes out the F1 score entirely a data problem more than a model problem.

Our confusion analysis reveals that errors are structurally informative: they almost always occur between signs that are visually similar within the same Gardiner group, suggesting that sequence-level or spatial context is the most productive direction for further accuracy gains. Future work will focus on:

- Expanding image counts for rare classes through new digitisation or synthetic generation [11, 17], targeting the full 750+ Gardiner sign list.
- Exploring attention-based and vision-transformer architectures for within-group confusion between visually similar signs.
- Training on more stylistically diverse data [13] to address cross-style generalisation.
- Integrating the classifier into downstream translation [3, 19] and retrieval-augmented pipelines [8], with broader NLP benchmarking [4] providing complementary insights.

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References

- [1] Ahmed Agha, Mohamed Sadek, Mohamed Badreldin, and Mohamed Ghoneim. 2018. *Dua-Khety: Hieroglyphics Detection, Classification and Transliteration*. Technical Report. Department of Computer Science and Engineering, American University in Cairo, Cairo, Egypt.
- [2] Andrea Barucci, Costanza Cucci, Massimiliano Franci, Marco Loschiavo, and Fabrizio Argenti. 2021. A Deep Learning Approach to Ancient Egyptian Hieroglyphs Classification. *IEEE Access* 9 (2021), 123438–123447. doi:10.1109/ACCESS.2021.3110082
- [3] Mattia De Cao, Nicola De Cao, Angelo Colonna, and Alessandro Lenci. 2024. Deep Learning Meets Egyptology: A Hieroglyphic Transformer for Translating Ancient Egyptian. In *Proceedings of the 1st Workshop on Machine Learning for Ancient Languages (ML4AL 2024)*. Association for Computational Linguistics, Bangkok, Thailand, 71–86. doi:10.18653/v1/2024.ml4al-1.9
- [4] Danlu Chen, Freda Shi, Aditi Agarwal, Jacobo Myerston, and Taylor Berg-Kirkpatrick. 2024. LogogramNLP: Comparing Visual and Textual Representations of Ancient Logographic Writing Systems for NLP. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, 14148–14167. <https://aclanthology.org/2024.acl-long.768/>
- [5] Costanza Cucci, Andrea Barucci, Fabio Nesi, Vittoria Del Vecchio, Marcello Piccollo, Lorenzo Stefani, Maria Messineo, and Fabrizio Argenti. 2025. AI-Based Methods and Hyperspectral Imaging Techniques Towards an Automated Translation of Ancient Egyptian Hieroglyphs: Perspectives and Open Problems. In *Optics for Arts, Architecture, and Archaeology (O3A) X (Proceedings of SPIE, Vol. 13569)*, Haida Liang and Roger Groves (Eds.). SPIE, Munich, Germany, 135690P. doi:10.1117/12.3066645
- [6] Costanza Cucci, Tommaso Guidi, Marcello Piccollo, Lorenzo Stefani, Lorenzo Python, Fabrizio Argenti, and Andrea Barucci. 2024. Hyperspectral Imaging and Convolutional Neural Networks for Augmented Documentation of Ancient Egyptian Artefacts. *Heritage Science* 12, 1 (2024), 75. doi:10.1186/s40494-024-01182-9
- [7] Jaime Duque-Domingo, Pedro Javier Herrera, Enrique Valero, and Carlos Cerada. 2017. Deciphering Egyptian Hieroglyphs: Towards a New Strategy for Navigation in Museums. *Sensors* 17, 3 (2017), 589. doi:10.3390/s17030589
- [8] Hesham A. Ebaid, Eyad Amgad, Zeyad Raslan, Ahmed Elkhoully, Khaled Behairy, and Ahmed Fares. 2025. Context-Aware Translation of Egyptian Hieroglyphs via Symbol Segmentation, Classification, and Retrieval-Augmented LLMs. In *Proceedings of the 2025 13th International Japan-Africa Conference on Electronics, Communications, and Computations (JAC-ECC)*. IEEE, IEEE, 7–12. doi:10.1109/JAC-ECC67970.2025.11417496
- [9] Rubén Fuentes-Ferrer, Jaime Duque-Domingo, and Pedro Javier Herrera. 2025. Recognition of Egyptian Hieroglyphic Texts Through Focused Generic Segmentation and Cross-Validation Voting. *Applied Soft Computing* 171 (2025), 112793. doi:10.1016/j.asoc.2025.112793
- [10] David Fuentes-Jimenez, Daniel Pizarro, Álvaro Hernández, Adin Bartoli, Laura de Diego-Otón, Sira Palazuelos-Cagigas, César Guerra Méndez, and Carlos Gracia Zamacona. 2025. The OCR-PT-CT Project: Semi-Automatic Recognition of Ancient Egyptian Hieroglyphs Based on Metric Learning. *arXiv preprint arXiv:2512.24197* (2025). doi:10.48550/arXiv.2512.24197
- [11] Shiqi Gao, Bo Hui, and Wei Li. 2024. Image Generation of Egyptian Hieroglyphs. In *Proceedings of the 2024 16th International Conference on Machine Learning and Computing (ICMLC)*. Association for Computing Machinery, Shenzhen, China, 389–397. doi:10.1145/3651671.3651771
- [12] Maksim Golyadkin, Valeria Rubanova, Aleksandr Utkov, Dmitry Nikolotov, and Ilya Makarov. 2025. Evaluation of Egyptian Hieroglyph Classification Across Diverse Writing Styles. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*. Association for Computing Machinery, Dublin, Ireland, 5336–5344. doi:10.1145/3746027.3754797
- [13] Maksim Golyadkin, Valeria Rubanova, Aleksandr Utkov, Dmitry Nikolotov, and Ilya Makarov. 2025. MEH: A Multi-Style Dataset and Toolkit for Advancing Egyptian Hieroglyph Recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 24488–24496. doi:10.1109/ICCV51701.2025.02270
- [14] Maiken Mosleth King. 2024. Towards a Pedagogy of Ancient Egyptian Language and Hieroglyphs. *Cahiers du CLSL* 68 (2024), 63–90. doi:10.26034/la.cdclsl.2024.6143
- [15] Luca Lombardi, Francesco Mercaldo, and Antonella Santone. 2024. Egyptian Hieroglyphs Localisation Through Object Detection. In *Proceedings of the 19th International Conference on Software Technologies (ICSOFT)*. SCITEPRESS, Dijon, France, 434–441. doi:10.5220/0012787900003753
- [16] Ahmed Nasser, Marwan Mohamed, Alaa Sherif, Basmala Mahmoud, Shereen Yehia, Asmaa Saad, Mariam S. El-Rahmany, and Ensaf H. Mohamed. 2025. HieroGlyphTranslator: Automatic Recognition and Translation of Egyptian Hieroglyphs to English. *arXiv preprint arXiv:2512.03817* (2025). doi:10.48550/arXiv.2512.03817
- [17] Ricardo Muñoz Sánchez. 2024. When Hieroglyphs Meet Technology: A Linguistic Journey Through Ancient Egypt Using Natural Language Processing. In *Proceedings of the Third Workshop on Language Technologies for Historical and Ancient Languages (LT4HALA) @ LREC-COLING 2024*. ELRA and ICCL, Torino, Italy, 156–169. doi:10.63317/2mhka44cjoqv
- [18] Waqas Bin Umer. 2024. *Translating Egyptian Hieroglyphs Using Deep Learning*. MSc thesis. National College of Ireland, Dublin, Ireland.
- [19] Philipp Wiesenbach and Stefan Riezler. 2019. Multi-Task Modeling of Phonographic Languages: Translating Middle Egyptian Hieroglyphs. In *Proceedings of the 16th International Conference on Spoken Language Translation (IWSLT)*. Association for Computational Linguistics, Hong Kong, Article 2019.iwslt-1.33. <https://aclanthology.org/2019.iwslt-1.33/>